

FICTIONAL DEMO SUBMISSION

For LMS grading-workflow demonstration only

Assignment 3 - Exponential, Logarithmic and Trigonometric Applications

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Selected responses

These original responses are intentionally concise and are provided only to demonstrate Moodle submission, download, and grading workflows.

Question 1

$A(6) = 1800(1.045)^6 = 2344.05$. For the target, $2500/1800 = 1.045^t$, so $t = \ln(2500/1800)/\ln(1.045) = 7.47$ years. The balance first exceeds \$2500 after about 7.5 years.

Question 2

The domain is $x > 1$. Combining logs gives $\log_2(x^2 - 1) = 3$, so $x^2 - 1 = 8$ and $x = \pm 3$. Only $x = 3$ is in the domain.

Question 4

Amplitude 3, period π , shift $\pi/4$ right, midline $y = -1$, maximum 2, and minimum -4. One cycle runs from $x = \pi/4$ to $x = 5\pi/4$.

Question 6

Exponential models are strong for proportional change but can become unrealistic when resources are limited. Sinusoidal models show repeating seasonal patterns but assume a stable period and amplitude.